

Topics of today's class

Last class:

- Chapter 9
 - Consumer's consumption/savings decision
 - Responses of consumer to changes in income and interest rates.

Today's class:

- Chapter 9
 - Government budget deficits
 - The Ricardian Equivalence Theorem.

Chapter 9

A Two-Period Model: The Consumption-Savings Decision and Government Deficits

Government Budget Constraints

First period budget constraint:

$$G = T + B$$

- G : government expenditures
- T : lump-sum taxes
- B : government debt: or “treasury security” in the US

Second Period budget constraint:

$$G' + (1 + r)B = T'$$

- G', T' : government expenditures and lump-taxes in period 2
- $(1 + r)B$: government must pay off the debt in period 2

Government Budget Constraints

Combining

$$G = T + B$$

$$G' + (1 + r)B = T'$$

Present value budget constraint:

$$G + G'/(1 + r) = T + T'/(1 + r)$$

Interpretation:

- In the long run, all government purchases must be paid for by taxes
- Present value of government purchases = present value of taxes

Government Budget Constraints

- There are N consumers in the economy each pay current tax of t , so that

$$T = tN$$

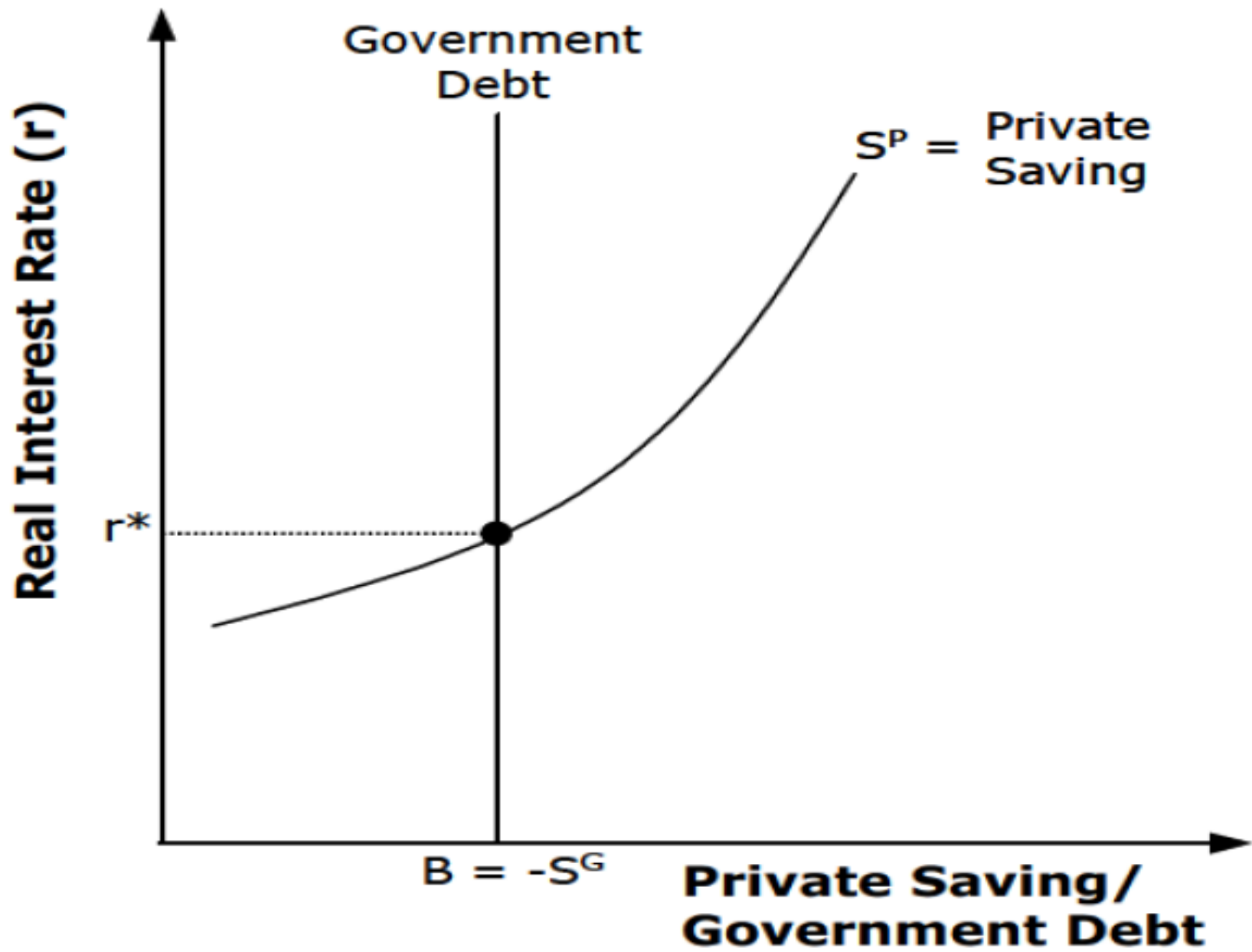
- In Period 2, consumers pay

$$T' = t'N$$

Credit Market Equilibrium

- Total private savings is equal to the quantity of government bonds issued in the current period.

$$S^P = B$$



Competitive Equilibrium

3 conditions must hold:

- Given $r : c, c'$ and s are optimal quantities for the consumer
- The government present value budget constraint holds
- The credit market clears: $S^P = B$

Ricardian Equivalence Theorem

Theorem: Tax changes have no effects on consumption

Intuition: Consider a government that cut taxes today

- lower taxes today require higher taxes tomorrow to satisfy the government present value budget constraint
- a tax cut today has no effect on the present value of taxes and on the lifetime wealth of the consumer
- Consumer cares only about its lifetime wealth

Proof of Ricardian Equivalence Theorem

T and T' must satisfy:

$$G + G' / (1 + r) = T + T' / (1 + r)$$

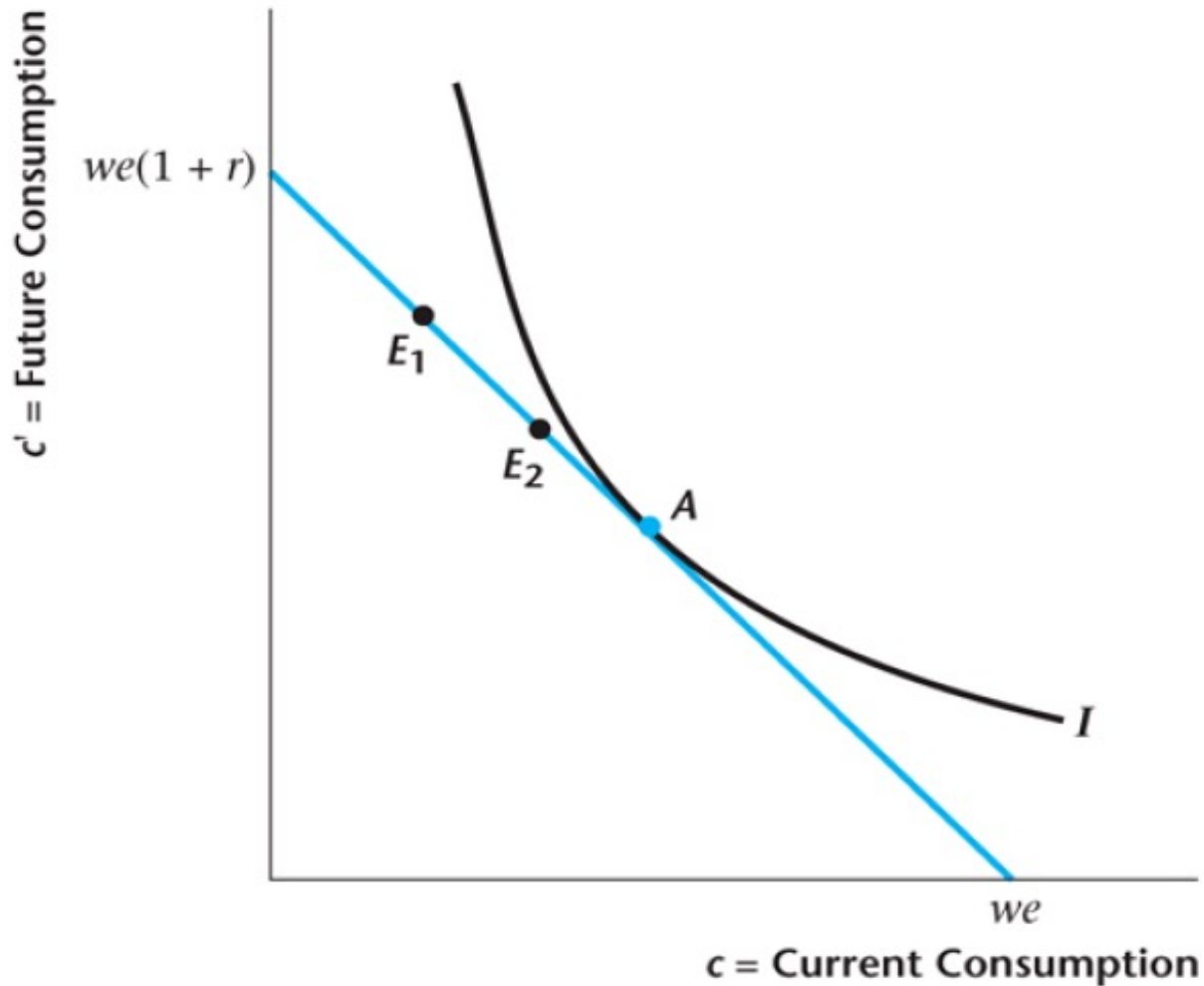
Any increase in T must be offset by an increase in T'

Using this result, rewrite the consumers lifetime wealth as:

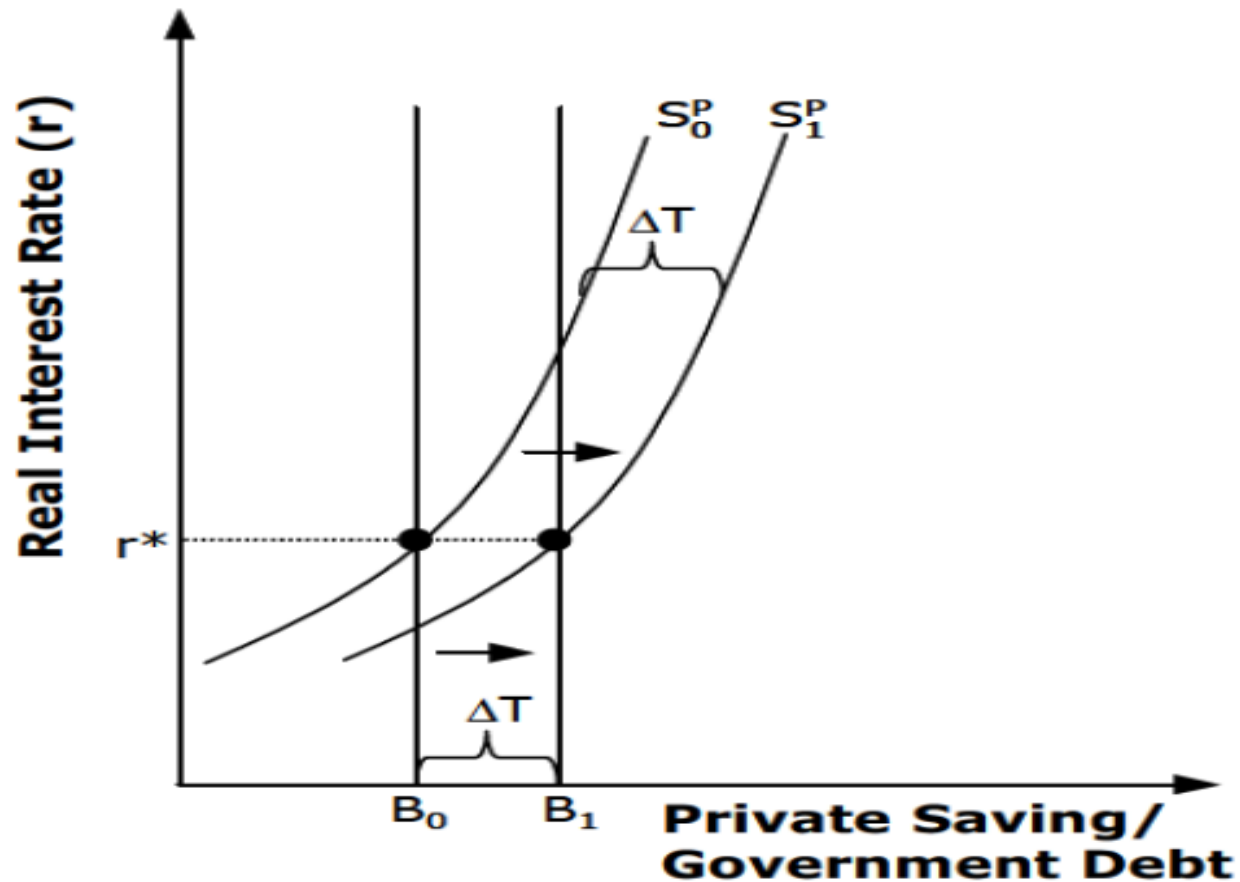
$$we = y + y' / (1+r) - (t + t' / (1+r))$$

to be continued....

Ricardian Equivalence – Consumer Decisions



Ricardian Equivalence – Consumer Decisions



Ricardian Equivalence Theorem

Ricardian equivalence shows that:

- tax cut today is not a free meal, it comes at a cost of higher taxes tomorrow
 - The government might put the tax increase off for two, three, or more years. Nevertheless, the current tax cut must be offset by future tax increases
- Under some conditions, rational agents should foresee that, and it should not have any welfare effects

Ricardian Equivalence Theorem

- The two more persuasive arguments against the Theorem are:
 1. finiteness of life
 2. imperfections of credit markets
- If tax cuts today, government can postpone the increase in taxes until long in the future: after the death of typical person alive during the tax cuts.
 - ➔ reduction in the present value of taxes
- Some of the burden of public debt is passed on to future generations

Government Deficits, Taxes and Government Debt

- If the U.S. government run a deficit of from 5% to 10% of GDP for a long period of time, what would be the consequences for the government debt?
- How would this matter in the future for the level of taxation faced by the average American?

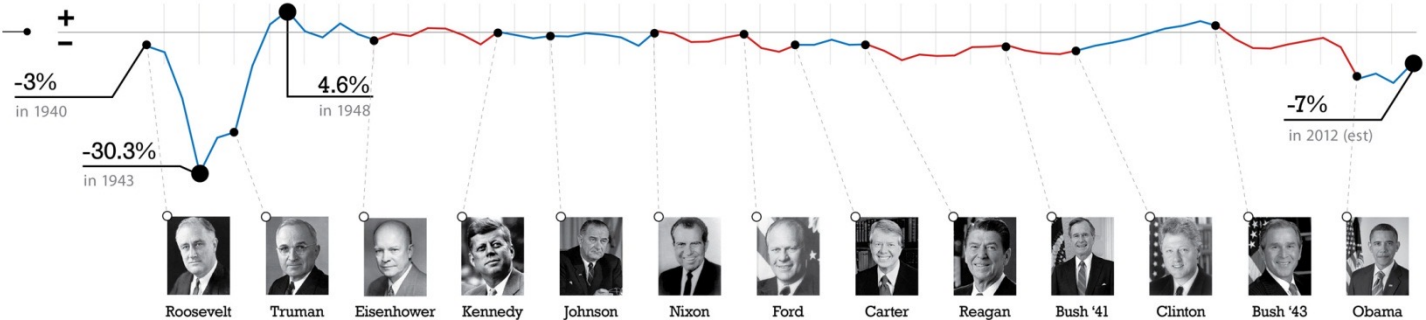
WHO IN WASHINGTON IS RESPONSIBLE FOR

THE U.S. DEBT

With all the talk lately about the debt ceiling, we wanted to take a look at how we got here. While political parties play the blame game, the facts paint a more complicated picture. The data shows that both parties have presided over huge increases and decreases in our national debt, and that events like World War 2 and periods of recession have often been far more important than party ideology.

BUDGET SURPLUSES & DEFICITS AS A PERCENTAGE OF GDP

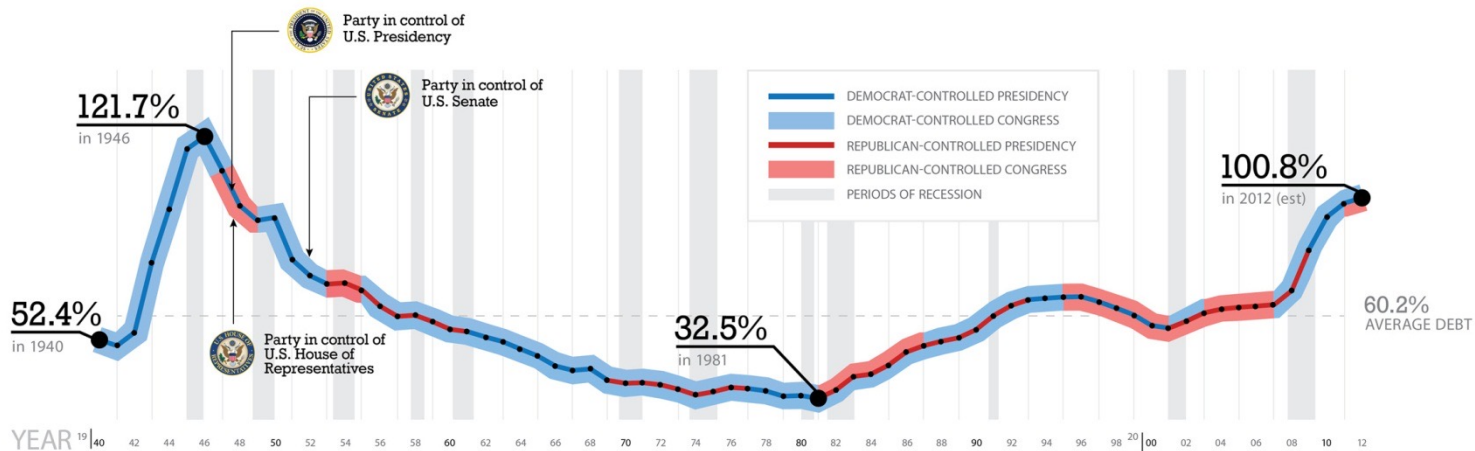
Some presidents are better at balancing budgets than others. Despite labels such as fiscal conservative or big government liberal, the data shows that the ability or inability to balance budgets is truly bi-partisan.



UNITED STATES DEBT AS A PERCENTAGE OF GDP (1940 - 2012 EST)

Measuring U.S. debt in numbers that haven't been adjusted for inflation produces an alarming and somewhat misleading result. $\frac{16.3}{.05}$ in 1940 (\$ trillions) in 2012

Measuring U.S. debt as a percentage of GDP gives us a much better idea of who our biggest borrowers have been.



HOW THE RATIO OF US DEBT COMPARES TO OTHER COUNTRIES.

The range is enormous due to emerging third world markets and wild swings created by the economic collapse. (Public Debt/GDP)



Government Deficits, Taxes and Government Debt

- Suppose that
 - GDP grows at a constant rate g ;
 - Government needs to finance a primary deficit of dY each period;
 - The market interest rate r is constant forever.

Government Deficits, Taxes and Government Debt

- In the long term, the government debt will be

$$B_t = dY_0[(1+r)^t + (1+r)^{t-1}(1+g) + \dots + (1+g)^t]$$

- The long run debt/GDP ratio will be

$$d\left(\frac{1+g}{g-r}\right)$$

- If $g=0.03$, $r=0.01$, and $d=5\%$, then debt/GDP ratio=2.575
- If $g=0.03$, $r=0.01$, and $d=10\%$, then debt/GDP ratio=5.15

Social Security

- Introduced after the Great Depression: Pay-As-You-Go Social Security:
 - currently working generation pays payroll taxes, whose proceeds are used to pay the pensions of the currently retired generation
 - current tax rate is 15.3% (you pay half and your employer pay the other half)
- Population Ageing:
 - more retirees to whom benefits are owed
 - fewer workers to pay for these benefits
- Every US Presidents (since Jimmy Carter) understood the problem but failed to reform the system

Social Security

- Overlapping Generations Model
 - Each agent lives two periods: Young and Old
 - In any period, there are people of different generations alive
- Population Evolution:

$$N' = N(1 + n)$$

Pay-As-You-Go Social Security

- Introduce Pay-As-You-Go Social Security:

- Old: Social security payments b
- Young: tax t

- Budget Equilibrium:

$$Nb = N't$$

- Total social security benefits must equal total taxes on the young

$$b / (1 + n) = t$$

- Social Security is a **forced savings program**.

Pay-As-You-Go Social Security

- When Pay-As-You-Go Social Security was introduced, The First Generation of Old Consumer clearly benefited from the program:
 - they did not pay taxes when they were young
 - they now have an additional source of income
- For everyone else, the effect on welfare is less obvious

Pay-As-You-Go Social Security

- Lifetime wealth of the consumer:

$$\begin{aligned} we &= y - t + \frac{y' - t'}{1 + r} \\ &= y - \frac{b}{1 + n} + \frac{y' + b}{1 + r} \\ &= y + \frac{y'}{1 + r} + \frac{b(n - r)}{(1 + r)(1 + n)} \end{aligned}$$

Pay-As-You-Go Social Security

- Lifetime wealth of the consumer:

$$we = y + \frac{y'}{1+r} + \frac{b(n-r)}{(1+r)(1+n)}$$

If $n > r$: Pay-As-You-Go Social Security improves welfare

If $n < r$: Pay-As-You-Go Social Security decreases welfare

Intuition:

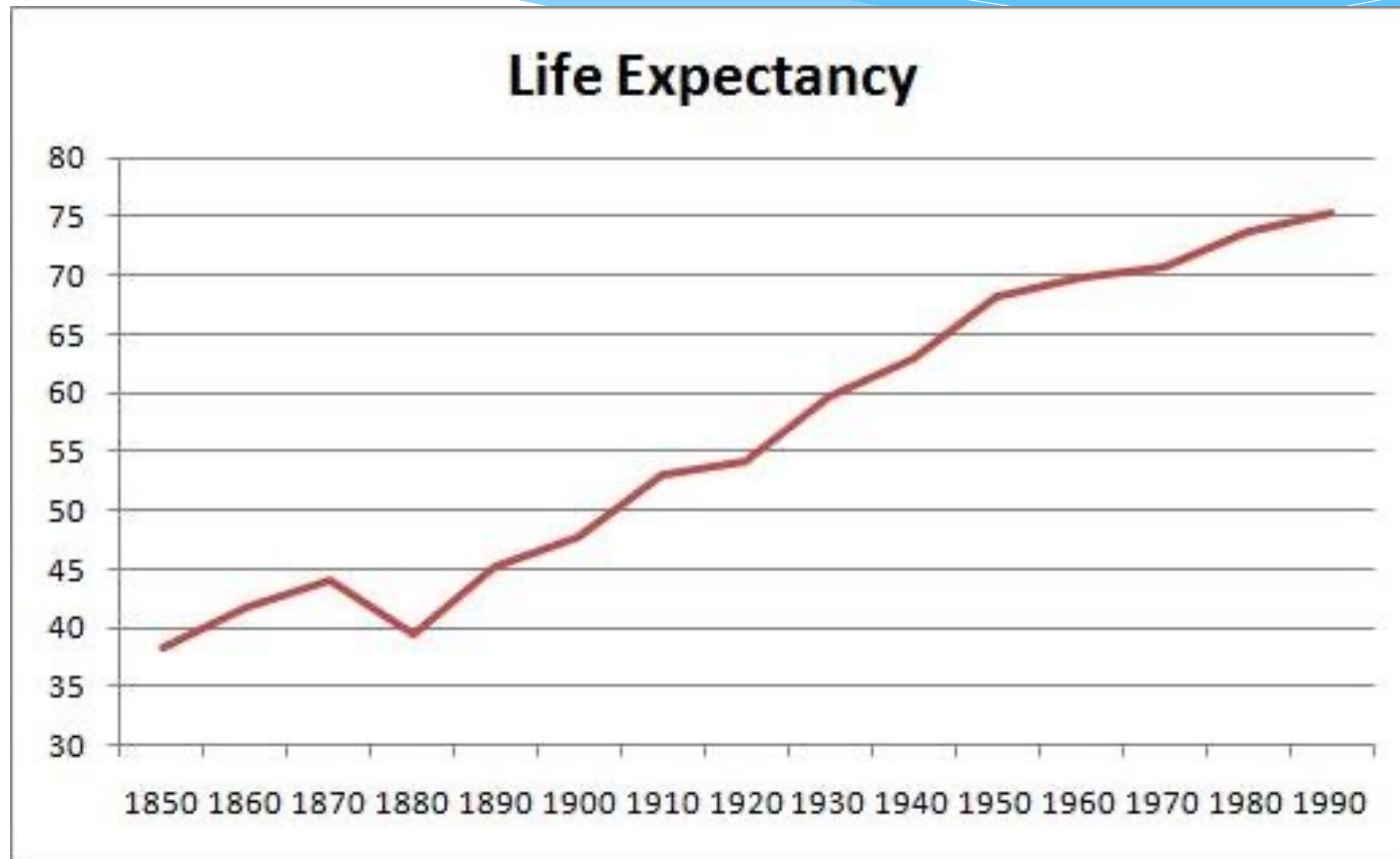
- If people save by themselves for retirement, return on their savings equals $1 + r$
- If people save via a social security system return equals:
 $1 + n$

Current Issues with Pay-As-You-Go Social Security

$$\text{payroll tax} = \frac{\text{pension income}}{\text{number of workers}} \times \frac{\text{number of retirees}}{\text{number of workers}}$$

- Population Ageing:
 - more retirees to whom benefits are owed
 - fewer workers to pay for these benefits
- Evolution of the Workforce:
 - 1940: ratio of workers to retirees=8
 - 2000: ratio of workers to retirees=4
 - 2040: ratio of workers to retirees=2
- But the problem is less severe than in most Europeans and Asian Countries

Social Security



Current Issues with Pay-As-You-Go Social Security

- Over the past 70 years, contribution of the worker is more than the benefits claimed of the retired
 - increase in the participation rate of women's, baby boomers' generation, strong growth, immigration
- This extra money has been placed in a trust fund that earns interest: \$2.5 Trillion (invested in US government securities)
- Social Security ran a surplus every year until 2021. Starting in 2021, Social Security's total cost exceeded its total income.
- Trust funds will be exhausted by 2035

Current Issues with Pay-As-You-Go Social Security

$$\text{payroll tax} = \frac{\text{pension}}{\text{income}} \times \frac{\text{number of retirees}}{\text{number of workers}}$$

Three main possibilities to balance the budget

1. raising retirement age

- because life expectancy is increasing
- with “hardship exemption”

Current Issues with Pay-As-You-Go Social Security

$$\text{payroll tax} = \frac{\text{pension}}{\text{income}} \times \frac{\text{number of retirees}}{\text{number of workers}}$$

Three main possibilities to balance the budget

2. reducing benefits

- decrease for the high levels of income
- increase for the poorest
- adjust the cost-of-living index

Current Issues with Pay-As-You-Go Social Security

$$\text{payroll tax} = \frac{\text{pension}}{\text{income}} \times \frac{\text{number of retirees}}{\text{number of workers}}$$

Three main possibilities to balance the budget

3. more progressive payroll taxes

- there is a maximum earnings level to which tax apply
- increasing this maximal taxable level

Social Security

Fully Funded Social Security :

- George W. Bush had proposed a transition to this system in 2005
- no taxes on the young and no transfers to the old
- The current worker's contribution is invested and used to pay their own benefits in the future

Two different methods:

- Privatization: putting workers' contribution into individual accounts and managed by individual workers
- Government can pool all the accounts and designate a management agency

Funds can be invested in

- government bonds only (safe!)
- diversified portfolio (but more financial risk!)

Transition to Full Funding

- There is one main problem: **Costly transition** from pay-as-you-go to full funding. **Missing generation:**
 - at the introduction of the system there was one generation that received social security but never paid taxes.
 - if pay-as-you-go disappears, there will be one generation that paid taxes but will not receive social security
- **Dilemma:**
 1. Currently young pay double
 2. Default on the promises for the old
 3. Increase government debt, financed by higher taxes in the future, i.e., by currently young and future generations. (Tax those who would benefit from switch.)